



Approvals



MUMBAI METRO RAIL CORPORATION LTD **(MMRCL)**

E-TENDER FOR

SUPPLY, INSTALLATION, TESTING AND COMMISSIONING OF
ELECTRICAL, AIR CONDITIONING, FIRE-ALARM WORKS AND OTHER
ALLIED SERVICES FOR MUMBAI METRO RAIL CORPORATION LTD.
PROJECT OFFICE AT E-BLOCK, BANDRA-KURLA COMPLEX, MUMBAI-
400 051

BID DOCUMENT

DATE: 15-12-2016.

MUMBAI METRO RAIL CORPORATION LIMITED
PLOT NO. C-14 & 15, 5th FLOOR A WING, OLD MMRDA BLDG.
BKC, BANDRA (E), MUMBAI, 400 051
Tel- 26597654, Email- ramesh.sharma@mmrcl.com
Website: www.mmrcl.com

8.11	SWITCHES	SEIMENS / ENGLISH ELECTRIC / L&T / C & S
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Sr. No	Component	Approved Make
8.12	STARTERS	SEIMENS / L&T / T. C.
8.13	PUSH-BUTTONS/OVERLOAD RELAY	SEIMENS / TECNIC / L&T / C & S / RASS
8.14	VOLTMETER / AMMETER	A/E - IMP
8.15	CUSHY FOOT MOUNTS	DUNLOP / EQVT. APPROVED
8.16	SIGNAL CABLES	RATRON / ASSOCIATED
8.17	POWER TRANSDUCERS	MECON / EQVT. APPRO.
8.18 8.19	P. V. C. PIPING CABLES	SUPREME / PRINCE / EQUIVALENT APPROVED (ISI)
	POWER CABLES CONTROL CABLES	ICC / CCI / GLOSTER FINOLEX / EQUI. APPROVED
8.20	COPPER PIPING	RAJCO/ MANDEV / PARASMANI

Note – The Vendor to highlight the make of material considered while quoting.



Approvals

PROJECT: 0.5 MTPA COPPER COMPLEX – HVAC WORK, MUNDRA

OWNER: KUTCH COPPER LTD

adani

adani

Categories

☒ **Doc I** Approved Used For Manufacturing/Construction/Installation

☐ **Doc II** Approved with minor comments. No re-submission is required. To be incorporated in the Bill. Used For Manufacturing/Construction/Installation

☐ **Doc III** Re-submission is required. Approved & Returned for Manufacturing/Construction/Installation subject to incorporation of comments.

☐ **Doc IV** Not Approved. Review & Resubmit For Approval

☐ **Doc V** For Information & Records

☐ **Doc VI** Incorporate Comments & resubmit for information & records

Note: Approval of this document does not absolve the Contractor/Supplier/Manufacturer from fulfilling Contractor's obligations in any way.

28-12-2023

CONTRACTOR: Shree HVAC Engineers



DOCUMENT NAME: TDS OF COPPER PIPING

DOC.NO. : Revision **1**

PROJECT DOC. TITLE: **HVAC SYSTEM FOR 0.5 MTPA COPPER COMPLEX - MUNDRA**

CONTRACTOR DOC. NO.: HVAC-1981-UTL-DE-TDS-032

Revision - **1**

APPROVAL STATUS : ISSUE FOR APPROVAL

1	27/12/2023								
0	26/12/2023								
		INITIAL S	SIGN	INITIALS	SIGN	INITIALS	SIGN	DETAILS OF REVISION	
		PREPARED		CHECKED		APPROVED			

TECHNICAL DATA SHEET				
SR.NO	CHEMICAL COMPOSITION / MECHANICAL PROPERTIES	JIS H3300	IS 2501	SUGGESTED MAKES
1	UNS NO.	C 1220	DHP IS 191 PART VIII	HARIOM, PARASMANI
2	Cu	99.9 MIN	99.9 MIN	HARIOM, PARASMANI
3	Fe	-	0.03	HARIOM, PARASMANI
4	P	0.015 - 0.040	0.015 - 0.010	HARIOM, PARASMANI
5	TEMPER CONDITION	O 1/2	OD	HARIOM, PARASMANI
6	TENSILE STRENGTH (Mpa) (SOFT)	206 MIN	-	HARIOM, PARASMANI
	TENSILE STRENGTH (Mpa) (HALF HARD)	245 - 324	22kg/sqmm MIN	HARIOM, PARASMANI
	TENSILE STRENGTH (Mpa) (HARD)	314 MAX	28kg/sqmm MIN	HARIOM, PARASMANI
7	ELONGATION (%)	40 MIN	40 MIN	HARIOM, PARASMANI
8	GRAIN SIZE	> 0.040	0.015 - 0.040	HARIOM, PARASMANI
		0.015 - 0.040		



Approvals



STATE BANK OF INDIA

INVITES E-TENDER

FOR

**PROPOSED AIR CONDITIONING WORK OF SBI SURAT MAIN BRANCH (00488), AT
CHAWK BAZAR, SURAT**

FROM

THE SBI AHMEDABAD EMPANELLED AIR CONDITIONING WORKS CONTACTOR
UNDER CATEGORY UP TO RS. 20.0 LAKH AS PER LIST ENCLOSED

THE LAST DATE OF SUBMISSION OF TENDER: 01.07.2021 UP TO 03:00 PM

PART - A: TECHNICAL BID

TENDER SUBMITTED BY:

NAME : _____

ADDRESS : _____

DATE : _____

LIST OF APPROVED MAKES - THE CONTRACTOR MUST USE THE MATERIAL FROM THIS LIST ONLY.						
NO	Component	proved Make - Contractor must highlight the product / make considered while quoting & also enclose detailed product catalogue for each type of machine as per BOQ with technical bid				
		h star rating as mentioned in the BOQ - The 1.0 Tr / 1.5 Tr / 2.0 Tr. Split AC units shall have certification from the Bureau of Energy Efficiency (BEE) 2019.				
1	A.C. Units (Split with copper condenser unit)	Carrier (Superia)	Daikin	Hitachi	Voltas	
2	Ductable & Cassette units	Carrier (Superia)	Daikin	Hitachi	Voltas	
3	VRF Systems	Carrier	Daikin	Hitachi	Toshiba	Mitsubishi Heavy
		Ogeneral	Mitsubishi Electric			
4	Expanded Poly-urethane	Owens Corning	Beard sell	Coolfine	Modiform	
5	Fiber glass / Glass wool	FGP	Kimmco	TWGA		
6	Wires - "FRLS"	FINOLEX	POLYCAB	R.R. CABLE	HAVELLS	CEI
7	Cables	FINOLEX	POLYCAB	R.R. CABLE	HAVELLS	CEI
8	Electrical Components / Switch gears	EE	L & T	SIEMENS		
9	Continuous grills & Diffusers	CARYAIRE	COSMOS / CHAUHAN	DYNACRAFT	AIR MASTER	
10	G.I. Sheets 120 GSM	ESSAR	JINDAL	TATA	SAIL	
11	Drain pipes - UPVC - SCH - 40	DUTRON	PRINCE	SUPREME	ASTRAL	PRAKASH
12	Electrical motor	ABB	CROMPTON	SIEMENS		
13	Ventilation fans	ABB / PUNKER	CENTIMASTER	COMEFRI	KRUGER/NICOTRA	
14	Air curtain	ACME	ADC	SAP		
15	Metering device	AE	IMP	MECO		
16	Indicating lamps	L & T	SIEMENS			
17	G.I. Pipe	GST	PRINCE	ZENITH		
18	Dampers	CARYAIRE	DYNACRAFT	RAVISTAR	AJANTA CITIZEN	
19	Copper pipe	MANDEV	TOTALLINE	PARASMANI	CITIZEN	MEXFLOW
20	STEEL for Outdoor stands	FRESH & GOOD QUALITY				
21	Armor	RAYNO	SUPREME	ROTOCAST	ESSELL	
22	M.S. Conduits & accessories Bends, Tees, Elbo Etc. shall be of the same make as the conduits	AKG	NAI	JMV	BEC	
23	Close cell Elastomeric Nitride rubber (sheet & tube)	K-FLEX / ARMACELL / ARMAFLEX	A-FLEX / VDOFLEX		SUPERLON	
24	Drain / Condensate pumps	ECKERLE / ASPEN	50 OR ABOVE (APPROPRIATE FOR THE TYPE & LOCATION OF MACHINE & AS PER SITE CONDITION) WITH ALL ACCESSORIES SUCH AS EXTENTION CABLE, PVC TUBE WITH CONNECTORS, INLINE FILTER, CHECK VALVE ETC. COMPLETE			
25	Anchor fastener	HILTI	FISCHER			
26	Acoustic insulation	UP TWGA	KRIMCO			
TES: THE CONTRACTOR HAS TO SUBMIT THE SAMPLES OF ALL ITEMS & GET THE APPROVAL FROM THE ARCHITECT BEFORE PLACING THE ORDER & BEFORE STARTING THE EXECUTION OF WORK						



Approvals



TENDER DOCUMENT

FOR

TOWARDS SUPPLY, INSTALLATION, TESTING AND COMMISSIONING OF VRF/VRV
AIR CONDITIONING SYSTEM ALONG WITH ASSOCIATED EQUIPMENT

COUNCIL OF SCIENTIFIC AND INDUSTRIAL RESEARCH
NATIONAL AEROSPACE LABORATORIES
PB NO.1779, HAL AIRPORT ROAD, BENGALURU - 560017
KARNATAKA-INDIA

Sl. No	Description	Unit	Qty	Rate in `	Amount in `
29.5	Supply air ducts including plenum with flanges and other fasteners of various sizes made out of GI sheet.	Sq.	180		
29.6	450 mm x 450 mm size Volume controlled supply air grills made out of MS sheet.	Nos.	30		
29.7	380 mm x 380 mm size Volume controlled supply air grills made out of MS sheet.	Nos	34		
29.8	4" MS B Class pipe, 4" Non Return Valve. and other accessories.	Lot	1		
Sub-total - E					
Abstract					
				Sub-total - A	
				Sub-total - B	
				Sub-total - C	
				Sub-total - D	
				Sub-total A + B + C + D	
				Sub-total - E	
				Grand total (Subtotal A + B + C + D) - (Subtotal E)	

LIST OF APPROVED MAKES OF MATERIALS

Sl. No	ITEM	APPROVED MAKE
1	VRF / VRV Systems	Mitsubishi / Hitachi / Trane / Toshiba / Dikin / Blue star / Voltas.
2	Indoors cassette type units	Mitsubishi / Hitachi / Trane / Toshiba / Dikin / Blue star / Voltas
3	Vibration pads	Resistoflex / Dunlop
4	Vibration isolation	Dunlop / Cushy Foot / Resistoflex
5	Refrigerant Pipes	Mandev / Parasmani / Jugal
6	Nitrile Rubber insulation	Hylam / Superlon / Armaflex / Trocellen
7	Power Cables	Universal / Nicco / Gloster / Torrent / Primecab / Polycab / Lapp India
8	Control Cables	Polycab / Finolex / Torrent / RR Kabel / LAPP
9	Panel builder	Load controls / Lotus power gear / Ellins switch board / ENCONPASS / Switchgear & control technique
10	Cable gland	Comex / Comet / Jainsan
11	Lugs / sockets	3D / Dawells / Jainsan
12	PVC pipe	Supreme / Finolex.

4.1.4 Scope of Supply includes the following:

- Supply of the equipment / system as specified in 4.1 above
- Installation & Commissioning as specified in 4.3.3 above
- Training as specified below in 4.4
- On site comprehensive Warranty for 2 Year (24 months) as specified in 4.5 (i) below
- 3 (Three) Year comprehensive maintenance contract after the expiry of 2 - years warranty



Approvals

Tender Notice No. NIDJ/TENDER/ESTATE/2020-21/2063
Dated: 18.06.2021



राष्ट्रीय डिज़ाइन संस्थान, असम
National Institute of Design, Assam

Tender Document

For

Supply, Installation, Testing and Commissioning of VRV/VRF HVAC System at Workshop Building of NID, Assam

National Institute of Design, Assam

Tocklai, Rajabari
Jorhat, Assam-785014

Sl. No	ITEM	APPROVED MAKE
1	VRF / VRV Systems	Daikin/Hitachi/Carrier/Voltas
2	Indoors units	Daikin/Hitachi/Carrier/Voltas
3	Vibration pads	Resistoflex /Dunlop
4	Vibration isolation	Dunlop / Cushy Foot /Resistoflex
5	Refrigerant Pipes	Mandev / Parasmani /Jugal
6	Nitrile Rubber insulation	Hylam / Superlon / Armaflex /Trocen
7	Power Cables	Universal / Nicco / Gloster / Torrent / Primecab / Polycab / Lapp India
8	Control Cables	Polycab / Finolex / Torrent / RR Kabel / LAPP
9	Panel builder	Load controls / Lotus power gear / Ellins switch board / ENCONPASS / Switchgear & control technique
10	Cable gland	Comex / Comet / Jainson
11	Lugs / sockets	3D / Dawells / Jainson
12	PVC pipe	Supreme / Finolex.



Approvals

Project Name: Chhattisgarh Inclusive Rural and Accelerated
Agriculture Growth (CHIRAG)

Procurement of Heating, Ventilation and Air Conditioning
Goods and Installation services for CHIRAG SPMU Office at
CBD Building, 8th Floor, Nawa Raipur, Chhattisgarh
Through
Request For Quotation (RFQ)/Shopping Procedure

DEPARTMENT OF AGRICULTURE DEVELOPMENT, FARMERS
WELFARE AND BIOTECHNOLOGY, GOVERNMENT OF
CHHATTISGARH

Annexure 3

Approved Make list

LIST OF APPROVED MAKES OF EQUIPMENTS / MATERIALS FOR AIR-CONDITIONING SYSTEM

1.	VRF/VRV AIRCONDITIONING Units	OGENERAL / MITSUBISHI / VOLTAS
2.	Air Handling Unit	System air / Crystal/Edgtech/ZECO
3.	AHU Fans	Comferi / Kruger / Green heck /System air / Airflow / Nicotra
4.	Packaged Type Scrubber / Air Washer	System air / Crystal/Edgtech/ZECO
5.	Scrubber & Air-washer Fans	Comferi / Kruger / Green heck /System air / Airflow / Nicotra
6.	Axial Fan / Centrifugal Fan	System Air / Kruger / Airflow / Flaktwoods / Humidin /Crystal/ Caryaire
7.	Three phase motors	ABB / CG / Siemens / Bharat Bijli / As per OEM of Major Equipment
8.	Water Piping	Tata / SAIL / Jindal / Surya
9.	Y-strainer / Pot – strainer	SKS/ CIM /Emerald / Flowel
10.	Butterfly Valve	Advance / SKS / EMERALD
11.	GSS Sheet	Tata / SAIL / Jindal / National / Bhushan
12.	Fire & Smoke Damper Spring Type	Ruskin(Titus) / Trox / Greenheck / Ravistar/Mapro / Systemair / Airflow
13.	Fire Damper Actuator	Belimo / Siemens / Danfoss / Eq. make
14.	Extruded aluminum grills / Diffusers / Dampers	Caryaire / Mapro / Ruskin/ Titus /Dynamic/Flowel / Systemair / Airflow
15.	Closed Cell Fire Retardant XLPE	Thermoflex / Supreme /K Flex / Cani / Armacell
16.	Nitrile Rubber Insulation	Vidoflex / K-flex / Eurobatex /Armacell / Supreme
17.	Copper Refrigerant Piping	Totalline/ Diamond / Star / Mexflow / Rajco / Parasmani



Approvals

Project: Centralized air conditioning work for Auditorium in Architecture building of SPA at NEW DELHI-110002.



SCHOOL OF PLANNING AND ARCHITECTURE

An 'Institution of National Importance' under the Act of Parliament

(Ministry of Human Resource Development, Government of India)

4, Block-B, Indraprastha Estate, New Delhi-110002

Tel:- 23724249

Fax:- 011-23702383

SECTION 9

LIST OF APPROVED MAKES

Sr.No.	Details of the Items	Manufacturer's name
1. Samsung	VRV System	Daikin / Mitsubishi / Toshiba / Panasonic /
2.	Inline Fans	Kanalfakt / Ostberg / Caryaire
3.	Propeller Fans	GEC Alstom / Khaitan / Crompton
4. Bijlee	Electric Motors	ABB / Crompton / Kirloskar / Siemens / Bharat
5.	CPVC Pipes	Astral
6.	GI Sheet	Sail /TATA / Bhushan
7.	Grilles/Diffuser	Caryaire / Ravistar / Mapro /Tristar
8.	MS Dampers /Louvres	Tristar / Mapro / Servex
9.	Control Cables	Grandlay /Batra Henlay /Kalinga
10.	Power Cable	ICC / Skytone / Polycab / Kalinga
11.	Nitrile rubber Insulation for pipe insulation	Eurabatax / Aeroflex / Totaline
12.	Flexible Duct Connection	Airflow / Pyroguard
13.	Gaskets	
14.	Adhesives	Fevicol / Superlon
15.	Vibration isolator	Resistoflex / Dunlop
16.	Filters	Thermadyne / Anfilco
17.	Refrigerant	Brassomatic / Totaline
18.	Polyethylene for duct insulation	Supreme / Paramount
19.	Fan Sections	Caryaire / Zeco / Vikram Hitech
20.	Refrigerant Pipes	Rajco / Parasmani / Shrishyam
21.	HRV/ERV units	DRI /Daikin / Mitsubishi / Toshiba / Panasonic

End of section